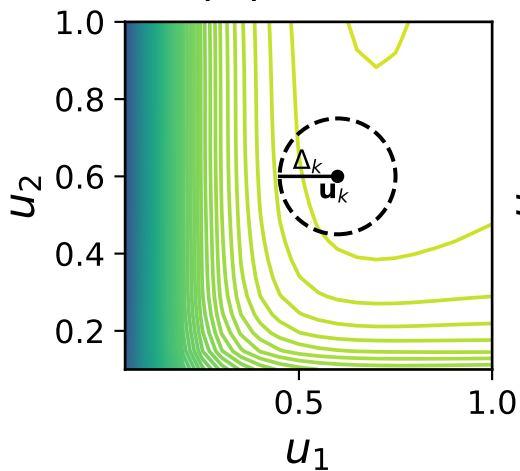
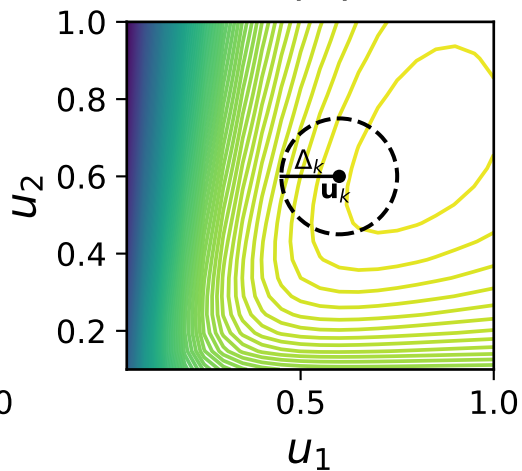
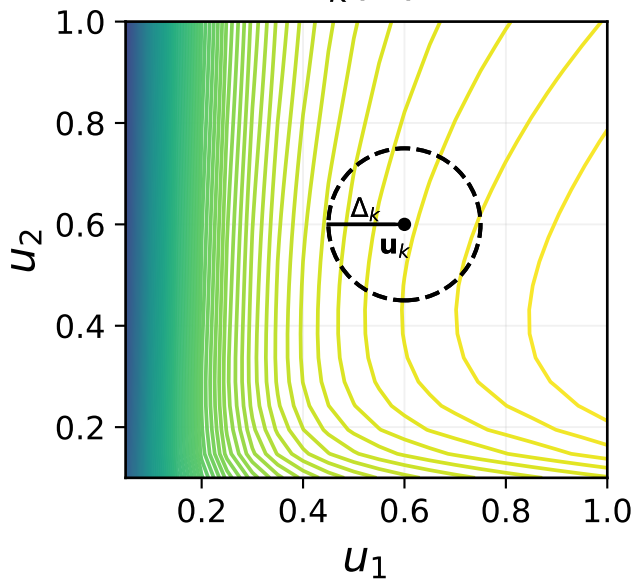


$\tilde{r}(\mathbf{u}) - \text{PINN}$  $s(\mathbf{u})$  $r_k(\mathbf{u})$ Oil production rate, $w_{o, \text{out}}$